Logs and Sysmon Ingestion

By Ponomareva Olesya

1 Introduction

Logs are fundamental to visibility, monitoring, and response.

They represent everything that happens in a **system, application, or network**, providing traceability into **what happened, when it happened, and how it happened.** In Windows environments, the **Event Log subsystem** is a primary source of this visibility.

2 Logs and Events

- **Log**: A timestamped record automatically generated by a system, application, service, or security tool.
- **Event**: A single instance of an action.
- **Log vs Event**: A log is the written record of an event.

3 Importance of Windows Event Logs

Windows Event Logs are one of the pillars of visibility in Microsoft environments.

They provide structured and searchable information on system, application, and security activities.

4 Categories of Windows Event Logs

4.1 Core Categories

- **System Logs**: Events logged by the operating system and its core components.
- **Application Logs**: Events related to installed applications and software.
- **Security Logs**: Authentication attempts, access control, and other security-related events.

4.2 Additional Categories

- **Setup Logs**: Events related to application or system installations.
- **Forwarded Events**: Logs collected from other systems in a centralized logging setup.

5 Accessing Windows Event Logs

Windows Event Logs are stored in `.evtx` files in:

`C:\Windows\System32\Winevt\Logs`

5.1 Methods of Access

1. **Using PowerShell**

   Example:

   ```
   Get-EventLog -LogName Application
   ```

   Note: To access Security Logs, PowerShell must be run as Administrator.

2. **Using the Event Viewer GUI**

   Open via `eventvwr.msc` to browse and filter logs.

6 Key Components of a Log Entry

Each log entry contains critical details for investigation:

- **Timestamp**: Exact date and time of the event. Accurate timestamps are critical for detecting, investigating and reconstructing incidents.
- **Event ID**: Numerical identifier for the event type.
- **Message/Description**: Human-readable explanation of what occurred.
- **Entry Type (Event Level):**
  - Information
  - Warning
  - Error
  - Critical
  - Success/Failure Audit

7 Event IDs: Identifying Activities

Event IDs allow quick identification of system activities and potential security issues.

7.1 User Account Activities

- 4720 — User account created.
- 4726 — User account deleted.
- 4741 — User account changed.

7.2 Logon Activities

- 4624 — Successful logon.
- 4625 — Failed logon attempt.

7.3 Remote Access and Scripting

- 4104 — Remote PowerShell command executed.

7.4 Process and Program Execution

- 4688 — New process created.
- 2003 — USB device connected.

7.5 Security and Audit Logs

- 1102 — Security log cleared.

8 Centralized Log Collection

When logs are collected across multiple systems, **centralized platforms** (e.g., SIEM solutions) enable:

- Scalable search and filtering
- Correlation of events across systems
- Faster detection and response

Examples: Splunk, Elastic (ELK/Elastic Security), Microsoft Sentinel, Wazuh.

9 Sysmon: Enhanced Windows Monitoring

Sysmon (System Monitor): an advanced Windows monitoring tool for collecting and recording detailed system activity.

9.1 Installing Sysmon

1. Download from Microsoft Sysinternals.
2. Extract the ZIP.
3. Install with:
   ```
   sysmon.exe -i -accepteula
   ```
4. Apply a configuration (e.g., SwiftOnSecurity's Sysmon config):
   ```
   sysmon -c sysmonconfig.xml
   ```

9.2 Viewing Sysmon Events

Navigate in Event Viewer:

```
Applications and Services Logs
          Microsoft
              Windows
                  Sysmon
                      Operational
```

10 Ingesting Sysmon into Wazuh

Wazuh uses the **Logcollector module** to collect logs from endpoints, applications, and network devices.

To integrate Sysmon logs into Wazuh:

1. **Update Wazuh Agent Configuration**

```
    <localfile>
      <location>Microsoft-Windows-Sysmon/Operational</
         location>
      <log_format>eventchannel</log_format>
    </localfile>
```

2. **Restart the Wazuh Agent**

```
    net start wazuh
```

3. **Verify Ingestion**

   - Check the Wazuh dashboard for Sysmon events.
   - Create custom decoders and rules for extracting fields.

11 Conclusion

Windows Event Logs are a cornerstone of visibility in Microsoft environments.

By enabling **centralized collection** and deploying **Sysmon**, organizations can significantly enhance detection and response.

When integrated into SIEM platforms like **Wazuh**, these logs become a powerful resource for identifying threats, investigating incidents, and strengthening defenses.